rewrite completely? Is there some way to convert the middleware to an endpoint?

Let's imagine that you need to apply authorization to the simple ping-pong endpoint you created in section 31.1.2. This is much easier to achieve with endpoint routing than simple middleware branches like `Map` or `Use`, but let's imagine you want to stick to using middleware instead of a traditional minimal API endpoint. The first step is creating a standalone middleware component for the functionality, using the approach you saw in section 31.1.4, as shown in the following listing.

Listing 31.9 The `PingPongMiddleware` implemented as a middleware component

```
public class PingPongMiddleware
{
    public PingPongMiddleware(RequestDelegate next)    ❶
    {
    }

    public async Task Invoke(HttpContext context)      ❷
    {
        context.Response.ContentType = "text/plain";   ❸
        await context.Response.WriteAsync("pong");     ❸
    }
}
```

❶ Even though it isn't used in this case, you must inject a RequestDelegate in the constructor.

❷ Invoke is called to execute the middleware.

❸ The middleware always returns a "pong" response.

Note that this middleware always returns a `"pong"` response regardless of the request URL; we will configure the `"/ping"` path later. We can use this class to convert a middleware pipeline from the branching version shown in figure 31.1, to the endpoint version shown in figure 31.4.